

 		LR2021RF2XDS1 Evaluation Module - 868/915 MHz & 2.4 GHz BOM			BOM # V1a	
		04.06.2026	Emr	Version e788v01a	SCH # e788v01a	PCB # e788v01a
RefDes	MPN	Geom	Value	Qty	Description	Manufacturer
U1	LR2021JMLTRT	QFN32L 5x5mm	LR2021	1	LR2021 LoRa Plus™ Low Power Multi Protocol Transceiver	Semtech
U2	320031231	Receptacle 1x7	7 pins	2	Receptacle for MCU XIAO Module - 7 Pins Through Hole Female Receptacle 5.00mm height, Top side	Dongguan Junliang Electronic Technology
U3	ZJY-2864KSWPG01	24.74 x 16.9mm	-	1	OLED Display Module with I2C Interface	ZJY Technology
U4	NCP59800BMNADJTAG	3mm x 3mm DFN8	8 pins	1	NCP59800 Adjustable Voltage Regulator - Low Noise, RF LDO - DFN8 0.65mm pitch	ON Semi
Resistors						
R3	CRCW0201120KFKE	0201	120k	1	Thick Film Resistor ±1%, 1/20W	Vishay/Dale
R4, N	ERT-JZER104F	0201	100k (nominal)	1	ERTJ Series Multilayer NTC Thermistor	Panasonic
R9, R13, R21	CRCW0201xxx	0201	220R	3	Thick Film Resistor ±5%, 1/20W (or equivalent)	Vishay/Dale
R7, R8, R11, R12, R15, R38, R39, R40, R45, R46, R47, R57, R58	CRCW0201xxx	0201	470R	13	Thick Film Resistor ±5%, 1/20W (or equivalent)	Vishay/Dale
R22, R31, R32, R33, R34, R35, R36, R37, R41, R42, R43, R44, R60, R61, R62, 868M, R915M, LR2021	CRCW0201xxx	0201	0R	18	Thick Film Resistor ±5%, 1/20W (or equivalent)	Vishay/Dale
R49, R50	CRCW0201xxx	0201	4.7k	2	Thick Film Resistor ±5%, 1/20W (or equivalent)	Vishay/Dale
R52	CRCW0402xxx	0402	470k	1	Thick Film Resistor ±5%, 1/16W (or equivalent)	Vishay/Dale
R51, R54	CRCW0402xxx	0402	0R	2	Thick Film Resistor ±5%, 1/16W (or equivalent)	Vishay/Dale
R53, R55, R56	CRCW0402xxx	0402	4.7k	3	Thick Film Resistor ±5%, 1/16W (or equivalent)	Vishay/Dale
R63	CRCW0402xxx	0201	33k	1	Thick Film Resistor ±1%, 1/20W (or equivalent) for 3.4VDC on U4 voltage output	Vishay/Dale
R64	CRCW0402xxx	0201	10k	1	Thick Film Resistor ±1%, 1/20W (or equivalent)	Vishay/Dale
R59, Other, LR2012, LR2022		0201	NC	0		Vishay/Dale
Capacitors						
C1, C18	GRM0335C1E470JA01D	0201	47pF	2	Multilayer ceramic capacitors COG ±5%, 25V	Murata
C2	GRM0335C1E220JA01D	0201	22pF	1	Multilayer ceramic capacitors COG ±5%, 25V	Murata
C3	GRM0335C1H7R5CA01D	0201	7.5pF	1	Multilayer ceramic capacitors COG ±0.25pF, 50V	Murata
C4	GRM0335C1H1R8BA01D	0201	1.8pF	1	Multilayer ceramic capacitors COG ±0.1pF, 50V	Murata
C7	GRM0335C1H2R2BA01D	0201	2.2pF	1	Multilayer ceramic capacitors COG ±0.1pF, 50V	Murata
C8, C12	GRM0335C1E180GA01D	0201	18pF	2	Multilayer ceramic capacitors COG ±2%, 50V	Murata
C9	GRM0335C1H3R9BA01D	0201	3.9pF	1	Multilayer ceramic capacitors COG ±0.1pF, 50V	Murata
C10	GRM0335C1H1R8BA01D	0201	1.8pF	1	Multilayer ceramic capacitors COG ±0.1pF, 50V	Murata
C13	GRM0335C1H6R6CA01D	0201	6.8pF	1	Multilayer ceramic capacitors COG ±0.25pF, 50V	Murata
C14	GRM0335C1H2R7BA01D	0201	2.7pF	1	Multilayer ceramic capacitors COG ±0.1pF, 50V	Murata
C15	GRM0335C1H1R1BA01D	0201	1.1pF	1	Multilayer ceramic capacitors COG ±0.1pF, 50V	Murata
C19	GRM0335C1H1R2BA01D	0201	1.2pF	1	Multilayer ceramic capacitors COG ±0.1pF, 50V	Murata
C20	GRM0335C1H2R0BA01D	0201	2.0pF	1	Multilayer ceramic capacitors COG ±0.1pF, 50V	Murata
C28	GRM0337R1C102KA01D	0201	1nF	1	Multilayer ceramic capacitors X7R ±10%, 16V	Murata
C56	GRM0337R1C103KE14D	0201	10nF	1	Multilayer ceramic capacitors X7R ±10%, 16V	Murata
C29, C33, C45	GRM155D80J25KE95D	0402	2.2uF	3	Multilayer ceramic capacitors X6T ±10%, 6.3V	Murata
C30, C34, C51, C52, C55, C57	GRM155R70J105KA12J	0402	1uF	6	Multilayer ceramic capacitors X7R ±10%, 6.3V	Murata
C31, C38, C39, C40, C42, C44, C46, C47, C48, C49, C54	GRM033C71C104KE14D	0201	100nF	11	Multilayer ceramic capacitors X7S ±10%, 16V	Murata
C32, C53, C58	GRM155C80J475MEAD0	0402	4.7uF	3	Multilayer ceramic capacitors X6S ±20%, 6.3V	Murata
C41, C50	GRM155R60J1106ME05J	0402	10uF	2	Multilayer ceramic capacitors X5R ±20%, 6.3V	Murata
C11, C27		0201	NC	0		Murata
Inductors & Ferrites						
L1	LQW15AN12N300D	0402	12nH	1	Wirewound Inductor ±5%	Murata
L2	LQP03HQ4N7H02D	0201	4.7nH	1	Thick Film Inductor ±3%	Murata
L3	LQP03HQ3N9B02D	0201	3.9nH	1	Thick Film Inductor ±0.1nH	Murata
L4	LQP03HQ24N4H02D	0201	24nH	1	Thick Film Inductor ±3%	Murata
L5	LQP03HQ7N4H02D	0201	7.5nH	1	Thick Film Inductor ±3%	Murata
L6	LQP03HQ2N4B02D	0201	2.4nH	1	Thick Film Inductor ±0.1nH	Murata
L7	LQP03TN18N302D	0201	18nH	1	Thick Film Inductor ±5%	Murata
L8	LQP03HQ1N5B02D	0201	1.5nH	1	Thick Film Inductor ±0.1nH	Murata
L9	LQP03HQ1N6B02D	0201	1.6nH	1	Thick Film Inductor ±0.1nH	Murata
L10	LQP03HQ1N1B02D	0201	1.1nH	1	Thick Film Inductor ±0.1nH	Murata
L11	LQM18PN2R2MGHD	0603	2.2uH	1	Chip Coil Inductor ±20%	Murata
L14	LQP03HQ22NH02D	0201	22nH	1	Thick Film Inductor ±3%	Murata
L15		0201	NC	0		Murata
FB1, FB2, FB4, FB5, FB6, FB10, FB15	BLM15PG100SN1D	0402	10R @100 MHz	7	Ferrite Bead 1A 25mohm	Murata
FB7, FB8, FB9, FB11, FB12, FB13, FB14	BLM03EB50SN1D	0201	50R @100 MHz	7	Ferrite Bead 400mA 50R @100 MHz	Murata
FB3, FB16	BLM15PG100SN1D	0402		0	Ferrite Bead 1A 25mohm	Murata
Diodes or LED						
D1	1N4148WT	SOD-523F	-	1	High Conductance Fast Switching Diode	ON Semi
D3, D4	RClamp2271ZC	DFN 0.60 x 0.30mm	-	0	RClamp2271ZC Ultra Small RailClamp® 1 Line, 22V EOS Protection ; recommended for RF antenna port	Semtech
Crystal or TCXO						
Q1	FTR5238-A0	QESM10 1.6x1.2mm	32.000MHz	1	Crystal unit; Tol. ±10ppm, Cload=10pF, G-Sensitivity=2ppb/G max (fitted)	Rakon
or	EXS00A-CS10665	NX1612SA 1.6x1.2mm	32.000MHz	0	Crystal unit; Tol. ±10ppm, Cload=10pF, G-Sensitivity=2ppb/G max (fitted)	NDK
or	FTR5123-A0	QESM09 2.0x1.6mm	32.000MHz	0	Crystal unit; Tol. ±10ppm, Cload=10pF, G-Sensitivity=2ppb/G max (2.0x1.6mm size !!!)	Rakon
or	EXS00A-CS06465	NX2016SA 2.0x1.6mm	32.000MHz	0	Crystal unit; Tol. ±10ppm, Cload=10pF, G-Sensitivity=2ppb/G max (2.0x1.6mm size !!!)	NDK
Connectors and others (or PIN equivalent)						
ANT, HF, ANT, LF	142-0711-841	SMA End Launch	50 Ohms	2	SMA End launch jack receptacle for PCB mount - for 1.20mm PCB Thickness	Cinch Connectivity Sol Johnson
IDCC	68000-102HLF	Header 1x2 2.54mm	2 pins	1	2 Pins - Through hole single row male connector, Top side	FCI
JH1, JH3	68000-108HLF	Header 1x8	8 pins	2	8 Pins - Through Hole Single Row Male Connector, Bottom side	FCI
JH2	68000-110HLF	Header 1x10	10 pins	1	10 Pins - Through Hole Single Row Male Connector, Bottom side	FCI
JH4	68000-106HLF	Header 1x6	6 pins	1	6 Pins - Through Hole Single Row Male Connector, Bottom side	FCI
J1, DISP, J3, DISP	75915-308LF	Receptacle 1x8	8 pins	0	8 Pins - Through hole single row female receptacle, Top side	FCI
J2, DISP	75915-310LF	Receptacle 1x10	10 pins	0	10 Pins - Through hole single row female receptacle, Top side	FCI
J4, DISP	75915-306LF	Receptacle 1x6	6 pins	0	6 Pins - Through hole single row female receptacle, Top side	FCI
J5, J6	1125S-4P	Header 1x4	4 pins	2	4 pins - Through Hole Male header Grove Connector with Lock, Ref Nr. 110990030 - Top side	Shenzhen NS-Tech
J7, SWD	95278-101-08LF	Header 2x8 SMD	8 pins	0	8 Pins - 2.54mm pitch - Surface Mount Dual Row Male Connector (or equivalent type)	Amphenol
JP1				0		
K1, USER, K2, RESET	TS-1111A	6.0x3.8mm	-	2	Normally Open Tactile Switch SMT without Positioning Drill Holes - TS-1111A Ref Nr. : 605C020333262	Xiamen Jinbeili Electronics
Additional or optional components						
(Jumper on IDCC)	M7566-05	Jumper 2.54mm	-	1	Red Jumper Socket 2.54mm on top of IDCC for DC supply current measurements	Hanwin
Shield		Shield	21.0x14.5mm	0	Custom Surface Mount RF-Shield in two parts with Cover (recommended for certification like FCC)	-

Remark: components not listed in the BOM or 0 quantity, are not populated on the module.